MIL-B-40042D

6 September 1974
SUPERSEDING
MIL-B-40042C
25 September 1970

MILITARY SPECIFICATION

BOX, MATCH, WATERPROOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a plastic waterproof matchbox with integrally mounted striking disc.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-349	-	Plastic Molding and Extrusion Material, Cellulose Acetate Butyrate.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

MILITARY

MIL-R-3065	-	Rubber, Fabricated Products.

STANDARDS

FEDERAL

FED-STD-406	-	Plastics: Methods of Testing.
FED-STD-595	-	Colors.

FSC 8465

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.
MIL-STD-417 - Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests).

DRAWINGS

ARMY NATICK LABORATORIES

8-2-10 - Box, Match, Waterproof.

(Figure 1 is a miniature reproduction of Drawing 8-2-10 and is attached for information only.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U.S. Postal Service Manual

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials Standard

D 787-70 - Ethyl Cellulose Molding and Extrusion Compounds

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

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Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Molding compound. The compound used in molding the box and cap shall be cellulose acetate butyrate conforming to type III, class MH of L-P-349 or ethyl cellulose conforming to type II, grade 7 of ASTM D 787-70.

* 3.1.2 Sealing gasket. The sealing gasket shall be made from rubber material meeting the requirements of type, class, grade and suffix letters RS 615A1Z of MIL-STD-417 and MIL-R-3065. Suffix letter "Z" indicates that there is a special requirement that the gasket material, in conjunction with the adhesive (3.1.4), striking disc (3.1.3), and the cap compound (3.1.1) shall be of a type which will permit the adhesion requirements of 3.3.2 to be met. The diameter of the gasket shall not be larger than the inside diameter of the cap and shall be not more than 1/64 inch smaller than the inside diameter of the cap.

3.1.3 Striking disc. The striking disc shall be commercial waterproof abrasive cloth, approximately 120 grit.

* 3.1.4 Adhesive. The adhesive or adhesives used for bonding shall bond the sealing gasket to the cap and the striking disc to the sealing gasket. The adhesive or adhesives shall be of a type and quality to meet the applicable performance requirements specified in 3.3.1, 3.3.2, and 3.3.3.

3.2 Design and construction. The matchbox shall conform to the details of design, construction and dimensions shown on Drawing 8-2-10.

3.3 Performance.

* 3.3.1 Waterproofness. The finished matchbox shall exhibit no penetration of water when tested as specified in 4.2.2.2.

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* 3.3.2 Adhesion of sealing gasket and striking disc. When tested as specified in 4.2.2.2, the sealing gasket shall not separate either partially or completely from the matchbox cap, and the striking disc shall not separate either partially or completely from the sealing gasket.

3.3.3 Dimensional stability. When tested as specified in 4.2.2.2, the matchbox shall not exhibit visual evidence of warpage and shall be free from cracks, swelling or distortion. The cap shall be capable of moving freely on its threads. The striking disc and sealing gasket shall not have loosened or changed position.

3.4 Color and luster. The color and luster of the matchbox shall approximate color No. 14087 of FED-STD-595.

* 3.5 Instruction sheet. Each matchbox shall be furnished with an instruction sheet inside. The instruction sheet shall consist of a piece of white paper 3 inches by 3 inches printed in bold faced type with the following legend:

"PLACE MATCHES IN MATCHBOX WITH THE HEAD OF THE MATCH TOWARD THE BOTTOM."

3.6 Marking for identification. Each matchbox shall have the letters "US" and the manufacturer's name or brand name embossed on the bottom as shown in Drawing 8-2-10.

3.7 Workmanship. The finished matchbox shall conform to the quality and grade of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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* 4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. In addition the supplier shall furnish a certificate of compliance stating that the striking disc conforms to the requirements specified in 3.1.3.

4.2.2 Inspection of end item.

4.2.2.1 Examination of the end item. The end item shall be examined for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.2.2.1.5. The lot size shall be expressed in units of matchboxes for examination in 4.2.2.1.1 and 4.2.2.1.2.

* 4.2.2.1.1 Visual examination. The end item shall be examined for the defects listed below. The sample unit shall be one matchbox.

Examine	Defect
Material:	
Molded plastic	Color and luster not as specified. Discolored, faded, resin-streak or mottled. Wrinkle, pimple or crater. Crazing, pitted, rough or porous surface. Surface scaly, hazy, chalky or burned. Any dirt in molding material. Any deep scratch or gouge mark on the exterior of the box.
NOTE:	Fine scratches may be disregarded.
Sealing gasket	Gasket not in one piece. Crack, extending through body. Cut, tear or hole. Bruise, loose skin, spongy, scar, void or otherwise defective.
Striking disc	Torn, area of grit missing, or otherwise damaged.

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Examine	Defect
Design and construction	Any part or component not fabricated of applicable referenced material, i.e., striking disc is not waterproof abrasive cloth.
	Varies from design, contour, or construction shown on specified drawing.
	Any characteristic not in accordance with specified requirements, i.e., diameter of gasket is larger than the inside diameter of the cap or diameter of gasket is more than 1/64 smaller than the inside diameter of the cap.
Workmanship and assembly	Threads missing or not type required.
	Threads broken, stripped, fractured or otherwise impaired.
	Flash on threads of box or cap.
	Cap does not move freely on its threads.
	Cap cannot be screwed on matchbox without pressure.
	Rough, sharp or broken edge on cap or body.
	Striking disc or sealing gasket detached.
	Adhesive extends beyond rim of striking disc.
	Adhesive tacky.
	Cap cannot be screwed on to give tight fit.
	Sealing gasket missing.
	Sealing gasket loose or changes position.
	Sealing gasket bulged out from too tight a fit.
	Clearance between bottom of cap and rim of body, when assembled with moderate force by hand, less than 1/64 inch.
	Striking disc missing or loose.
	Striking disc not positioned in center of gasket.
	Striking disc not positioned with the abrasive side showing.
	Any depression or void (other than superficial) on interior bottom of the matchbox.
	Surface and curves not blended smoothly.
	Body or cap warped, twisted, deformed, bent out of shape, split, or otherwise impaired.
Instruction sheet	Missing or not as specified.
	Incomplete or illegible.
Marking	Missing, incomplete, illegible, misspelled, not specified type, not in specified location, or not applied in the specified manner.

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4.2.2.1.2 Dimensional examination. Any dimension that is not within the specified tolerances shall be classified as a defect. The sample unit for this examination shall be one matchbox.

* 4.2.2.1.3 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing and marking comply with the section 5 requirements. Defects shall be in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level and AQL shall be as specified in 4.2.2.1.5.

Examine	Defect
Marking (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence or method of application.
Materials	Any component missing, damaged or not as specified.
Workmanship	Inadequate application of components such as: incomplete closure of container flaps, improper taping, loose strapping or inadequate stapling. Bulged or distorted container.
Contents	Number of matchboxes per interior package or container is more or less than required.

4.2.2.1.4 Examination of palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level and AQL shall be as specified in 4.2.2.1.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.

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Examine	Defect
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

* 4.2.2.1.5 Inspection levels and AQLs. The inspection levels and AQLs expressed in defects per hundred units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	II	6.5
4.2.2.1.2	S-3	2.5
4.2.2.1.3	S-2	2.5
4.2.2.1.4	S-1	6.5

* 4.2.2.2 Testing of the end item. The end item shall be tested for the characteristics listed in table I. The lot size shall be expressed in units of one matchbox. The sample unit shall be four finished matchboxes. The inspection level for each characteristic shall be S-1 and the AQL for each characteristic shall be 1.5 defects per 100 units.

TABLE I. End item tests

Characteristic	Requirement paragraph	Test method	No. determinations per sample unit	Results reported as
Waterproofness of matchbox	3.3.1	4.3.1	1	Pass or fail
Adhesion of sealing gasket:				
As is	3.3.2	4.3.2	1	Pass or fail
Oven aged	3.3.2	4.3.2	1	Pass or fail
After water immersion	3.3.2	4.3.2	1	Pass or fail
Adhesion of striking disc:				
As is	3.3.2	4.3.2	1	Pass or fail
Oven aged	3.3.2	4.3.2	1	Pass or fail
After water immersion	3.3.2	4.3.2	1	Pass or fail

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TABLE I. End item tests (cont'd)

Characteristic	Requirement paragraph	Test method	No. determinations per sample unit	Results reported as
Dimensional stability:				
After exposure at 140°F	3.3.3	4.3.3	1	Pass or fail
After exposure at -40°F	3.3.3	4.3.3	1	Pass or fail
Drop test	3.3.3	4.3.3	1	Pass or fail

4.3 Test methods.

* 4.3.1 Waterproofness of matchbox. The completely assembled matchbox shall be conditioned for 24 hours at 73.5° ± 2°F and 50 percent ± 4 percent relative humidity. Six wooden matches (nonsafety type) shall then be put in the box head down and the cap screwed on tightly by hand. The matchbox shall be submerged for 24 hours under 2 feet of tap water at room temperature (70° to 77°F). At the end of this period, the matchbox shall be removed from the water and dried with an absorbent material to remove surface moisture. The box shall be opened and each match shall be struck against the striking disc not more than ten times to test for ignitability. Failure of two or more matches to ignite shall constitute failure of the waterproofness of matchbox test.

* 4.3.2 Adhesion of sealing gasket and striking disc. A matchbox cap complete with sealing gasket and striking disc shall be subjected to four consecutive impacts of 12 inch-pounds each, as follows: The cap shall be placed open side down on a smooth flat steel base at least 4 by 8 by 3/4 inches thick. The cap shall be centered under a vertical tube which guides a dropping steel rod. The rod shall be 1 inch diameter and of a length to weigh 2.0 pounds, with the bottom end of the rod flat and squared to the centerline. The guide tube shall be not larger than 1-1/16 inch in diameter and shall permit an unobstructed free fall of the rod. A 1 inch diameter steel bearing ball shall be placed on the top surface of the cap and the guide tube positioned so as to keep the ball in a central position on the cap. The rod shall then be raised a height of 6 inches above the top of the ball and allowed to drop. The cap shall be impacted four times in this manner. The gasket and disc shall be examined visually for partial or complete separation from their positions. A separate

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matchbox cap complete with gasket and disc shall be tested in the same way after being subjected to oven aging at 140°F + 2°F for 24 hours and conditioned for 4 hours minimum under standard conditions. Separation of gasket or disc in either the oven-aged or non-oven-aged specimen will constitute failure in this test. The matchbox cap complete with sealing gasket and striking disc shall be immersed for 24 hours in tap water at 70° to 77°F. At the end of this period, the cap shall be removed and the gasket and disc examined visually for separation in whole or in part from their positions.

4.3.3 Dimensional stability. The completely assembled matchbox shall be conditioned for 24 hours at 73.5°F + 2°F and 50 percent + 4 percent relative humidity. The matchbox shall then be subjected to the accelerated service test in accordance with Method 6011 of FED-STD-406, except that the following exposure cycle periods shall be used:

Period of exposure (hours)	Temperature (°F)	Relative humidity (%)
24	140	85-89
24	-40	---

The matchbox shall be examined immediately at the end of each 24 hour period for visual evidence of warpage, cracks, loss in luster, distortion or swelling. After exposure to -40°F, the matchbox shall be dropped on a concrete floor or similarly hard surface from a minimum height of 3 feet. The striking disc and sealing gasket shall not become detached, loosened or cracked. It is not required that percentage changes in weight or dimensions be reported.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One hundred matchboxes, each with its cap turned finger tight, shall be packaged in a fiberboard box conforming to style RSC, type CF (variety SW), class domestic, grade 200 of PPP-B-636. The matchboxes shall be packaged upright 10 in length, 10 in width and 1 in depth within each fiberboard box. Inside dimensions of the box shall be 12 inches in length, 12 inches in width and 3 inches in depth. Box closure shall be secured with 2 inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

* 5.1.2 Level C (commercial packaging). Matchboxes shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The packages and quantity per package shall be the same as that normally used by the supplier for retail distribution.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

* 5.2.1 Level A. Six hundred matchboxes, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A packages shall be packed flat, two in length, one in width and three in depth within each shipping container. Inside dimensions of each shipping container shall approximate 25-1/4 inches in length, 12-3/4 inches in width and 11-1/4 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of the container specification.

5.2.2 Level B. Six hundred matchboxes, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A packages shall be packed flat, two in length, one in width and three in depth within each shipping container. Inside dimensions of each shipping container shall approximate 25-1/4 inches in length, 12-3/4 inches in width and 11-1/4 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification.

* 5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

* 5.2.3 Level C. Matchboxes, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with the U.S. Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 5.3 Palletization. When specified (see 6.2), matchboxes, packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

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5.4 Marking. In addition to any special marking required by the contract or order; interior packages, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The matchbox (with contents) is intended to be included in military personnel survival kits, with no limitations as to climatic conditions.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(c) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(d) When palletization is required (see 5.3).

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

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Review activity:

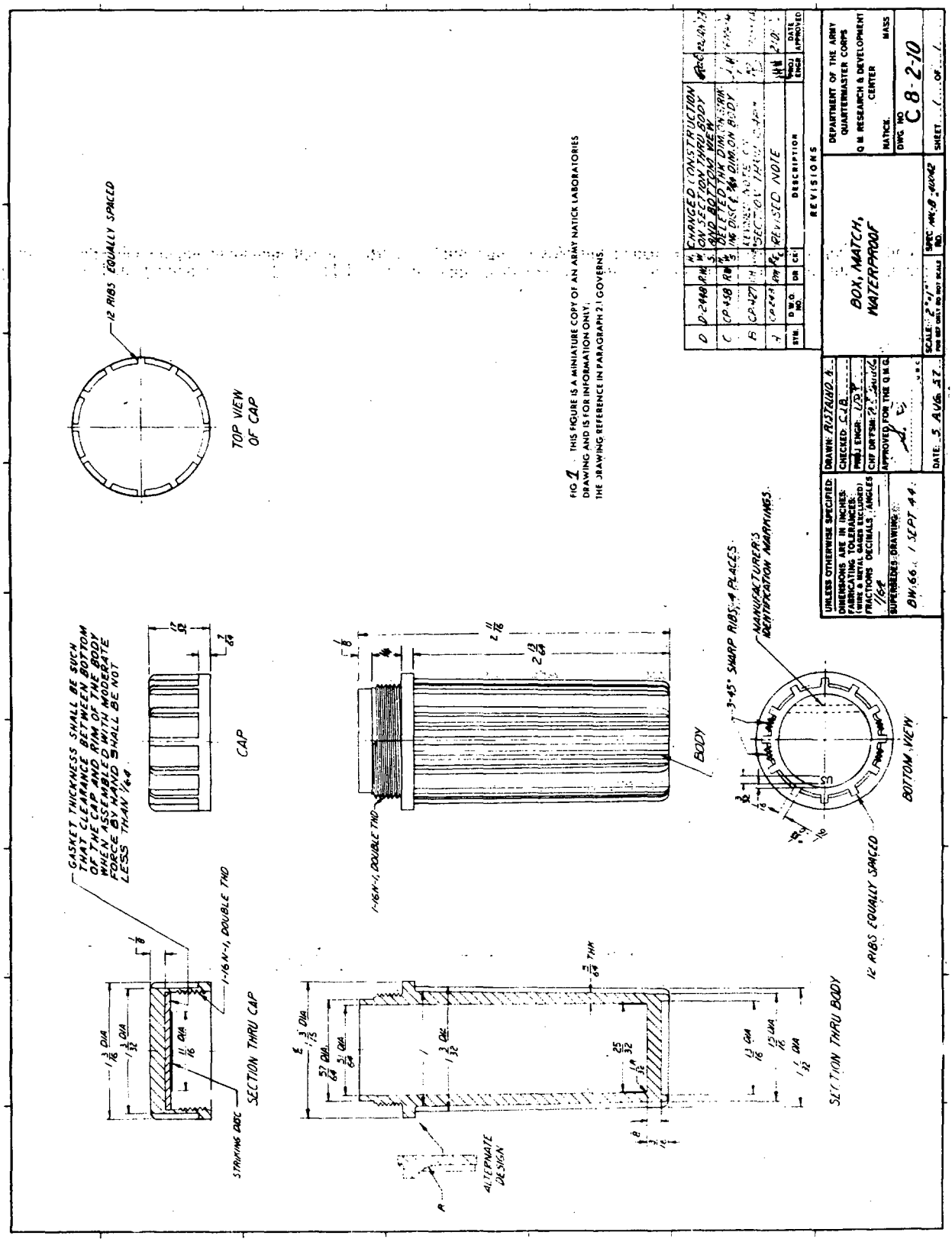
Navy - MS

Preparing activity:

Army - GL

Project No. 8465-0652

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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